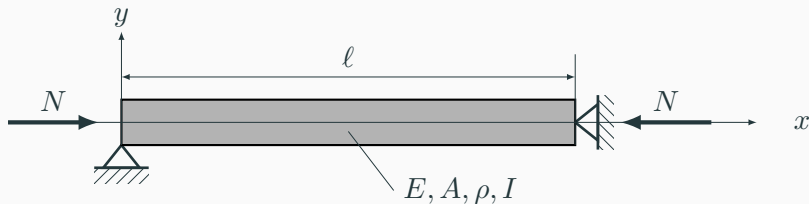


# Geometric stiffness

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## Beam subjected to axial loads

- Consider a beam subjected to axial forces  $N$  applied at both ends.

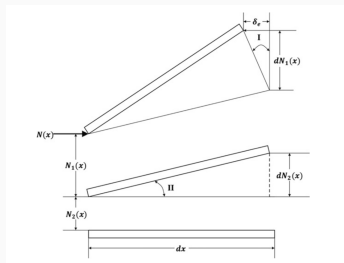


- Beam carrying large axial loads or undergoing large displacements have nonlinear behavior arising from the internal moments that are the product of the axial loads and the displacements transverse to the loads.
- The stiffness coefficients  $k_{ij}$  must be modified by the presence of the axial force to the corresponding **geometric stiffness coefficients**  $k_{ij}^g$ .

# Geometric stiffness matrix

$k_{ij}^g$  is defined as the force corresponding to the nodal coordinate  $i$  due to a unit displacement at coordinate  $j$  and resulting for the axial force  $N$ .

**Calculation of  $k_{12}^g$ :** vertical force at node 1 due to a unit rotation  $\theta_2 = 1$  caused by the axial force  $N$ . May be evaluated by the principle of virtual work:



$$dW = N\delta_e$$

By similar triangles we have

$$\frac{\delta_e}{dh_1(x)} = \frac{dh_2(x)}{dx}$$

$$\delta_e = \frac{dh_1}{dx}(x) \frac{dh_2}{dx}(x) dx$$

$$k_{12}^g = N \int_0^\ell h_1'(x) h_2'(x) dx$$

# Geometric stiffness matrix

- In general, any geometric stiffness coefficient may be expressed as

$$k_{ij}^g = N \int_0^\ell h'_i(x) h'_j(x) dx$$

- We define the **geometric stiffness matrix** as:

$$\mathbf{K}^g = N \int_0^\ell \frac{d\mathbf{H}^T}{dx} \frac{d\mathbf{H}}{dx} dx$$

- When  $N$  is constant along the beam length we obtain

$$\mathbf{K}^g = \frac{N}{30\ell} \begin{bmatrix} 36 & 3\ell & -36 & 3\ell \\ 3\ell & 4\ell^2 & -3\ell & -\ell^2 \\ -36 & -3\ell & 36 & -3\ell \\ 3\ell & -\ell^2 & -3\ell & 4\ell^2 \end{bmatrix}$$

# Example: stability of Bernoulli beam

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